

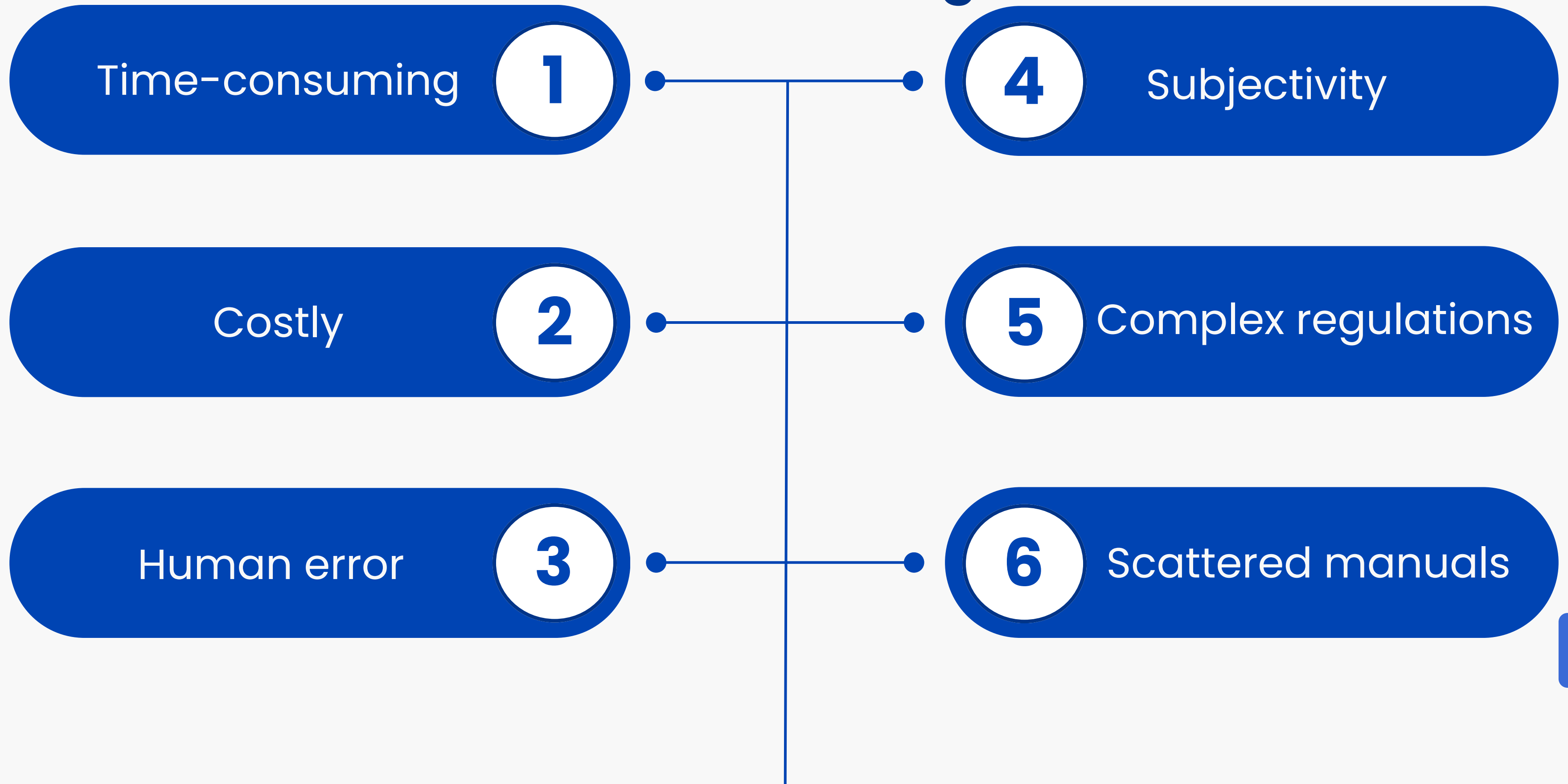
TEAM KORVAPUUSTI

AuditMax

Janction Hackathon 2025



The Problem With Auditing





Solution



Auto-ingests
regulations and
MOE manuals



Regulation
referencing



Generates question to
the MOE for the auditor



Highlights where
shortcoming are in the
MOE

Differentiators

Hybrid Compliance Engine

Hybrid Compliance Engine Combines traditional TF-IDF matching with AI-powered semantic search (Gemini) for regulation-to-MOE comparison. Enables robust cross-language and context-aware matching, outperforming simple keyword or rule-based systems.

Evidence Traceability

Extracts and stores character-level positions and page info for every finding, allowing users to trace results back to the exact source in the document. Improves audit transparency and trust compared to black-box solutions.

Modular & Scalable Architecture

Clean separation of frontend, backend, parsing, compliance logic, and database integration. Supports asynchronous job processing and real-time progress tracking for large audits.

End-to-End Automation

Fully automates the workflow: PDF upload → text extraction → section parsing → compliance assessment → result storage in Supabase. No manual mapping or pre-formatting required; supports real-world, unstructured documents.

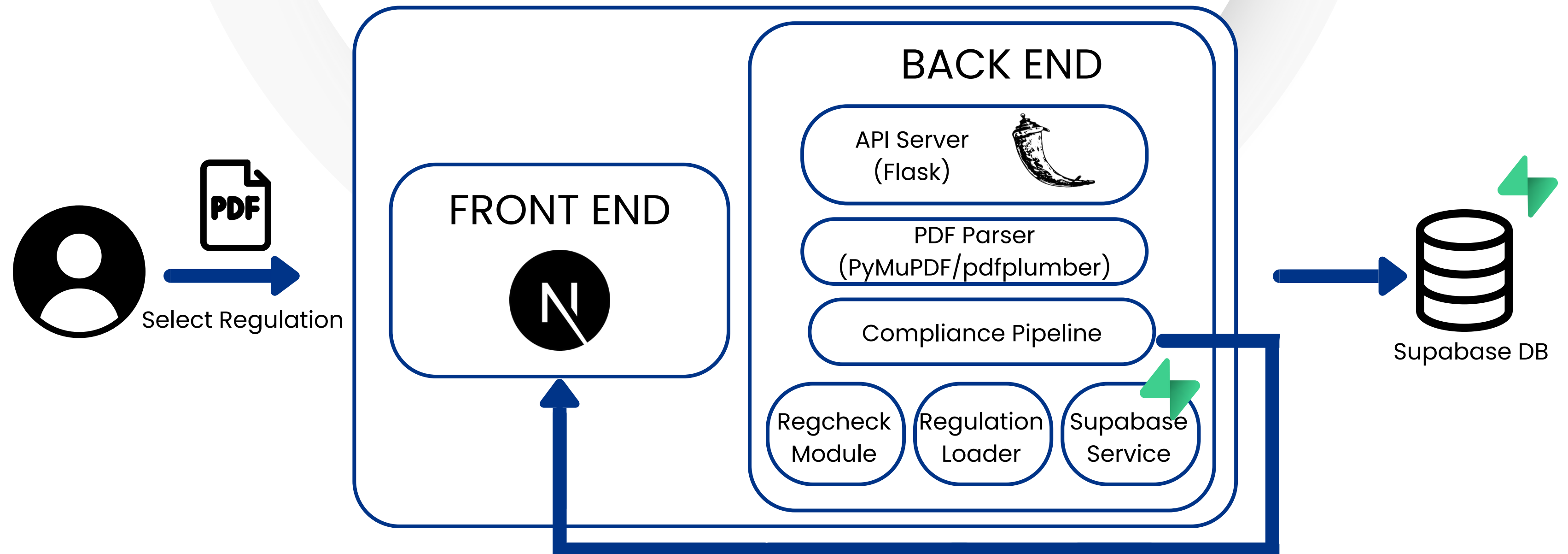
Domain-Specific Logic

Custom contradiction and gap detection logic tailored for aviation regulations, including time period, frequency, and requirement level checks.

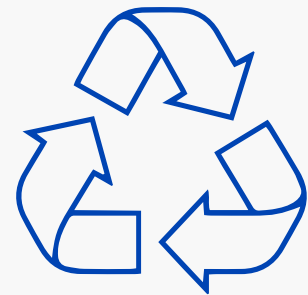
Accessible & User-Friendly UI

Uses Next.js and Radix UI for a modern, accessible frontend, including screen reader support and responsive dialogs.

System Architecture



Value Proposition



Reproducible

- System snapshots and caching store intermediate results
- Ensures AI outputs are identical on repeated runs
- Enables audit trail replay and regression testing
- Handles asynchronous long-running processes reliably



Secure

- All data stored in encrypted Supabase service-role database
- GDPR-compliant handling of MOE uploads and audit results
- Role-based access control and API authentication
- Audit logs for access and modification tracking



Transparent

- Each finding mapped to exact regulatory paragraph (EASA Part-145 / AMC / GM)
- Evidence pages and sentences highlighted for auditor review
- Scoring & gap detection logic explainable (TF-IDF + Gemini hybrid model)
- Supports claim-evidence traceability for regulatory audits

Evidence-Driven & Scalable



Multi-Sector Expansion

Aviation → Maritime → Rail → Road → Cybersecurity.
Same workflow, different rules.



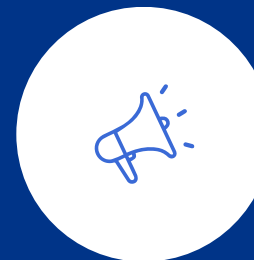
Future: Industry Test Bench

The same engine can become a “pre-check tool” for maintenance organizations to validate their manuals before submitting them



Clear traceability

Auditors can see where each finding comes from — no guessing, no black box



Support manual verification

The system generates simple, targeted questions based on the regulation, helping auditors confirm key points faster

Thank You



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